

PARTH DHARMENDRA PRAJAPATI

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EDUCATION

University of Southern California

Los Angeles, California

Master of Science in Computer Science (CGPA: 3.52 / 4.0)

August 2023 - May 2025

- Relevant Courses: AI, Machine Learning for Data Science, Analysis of Algorithm, Database Systems, Web Technologies

University of Mumbai

Mumbai, India

Bachelor of Engineering in Information Technology (CGPA: 9.19 / 10)

August 2019 - May 2023

- Relevant Courses: Data Structures and Algorithms, Web Development, Operating System, Big Data Analytics

SKILLS

- Programming languages:** Python, JavaScript, Java, C
- ML Libraries:** Scikit-learn, PyTorch, NumPy, Matplotlib, Pandas, Keras, TensorFlow, LangChain, TRL (Transformer Reinforcement Learning), Hugging Face Transformers
- Web Technologies:** React, Node.js, Express.js, AJAX, MongoDB, MySQL, HTML, CSS, Flask, Streamlit, AWS
- Soft Skills:** Problem Solving, Critical Thinking, Leadership, Teamwork, Communication, Time Management, Adaptability

PROFESSIONAL EXPERIENCE

Vayu Technology Corp.

Los Angeles, California

Data Science Intern

August 2024 – April 2025

- Designed and implemented algorithms for the Equilibrium System, analyzing large-scale sensor datasets to improve performance and reliability
- Developed production-ready logic for Pocket TUG (Timed Up and Go), pushup, endurance run, and walk features, identifying key patterns and optimizing detection accuracy
- Built a robust threshold-based model for calibration checks across multiple distributed body segment sensors, reducing error rates to less than 10%
- Created and refined prompts for an LLM application, optimizing responses and maintaining consistent output quality
- Collaborated in a cross-functional team, documenting designs and delivering maintainable Python code

University of Southern California

Los Angeles, California

Lead Course Producer (ISE 583: Enterprise-Wide Information Systems)

January 2025 – May 2025

- Led course logistics for ISE 583 course, coordinating other Course Producers and managing a class of 59 students with the professor
- Graded assignments and exams, provided timely feedback, conducted mentoring hours, and offered academic support
- Assisted in conducting the ERP Simulation Game, tested assignments, and clarified course material to enhance student learning

University of Southern California

Los Angeles, California

Teaching Assistant (ITP 259: Basics of Artificial Intelligence)

August 2024 – December 2024

- Assisted with grading assignments and exams for ITP 259, providing timely feedback to 30 students alongside the professor
- Supported students with course material clarification and conducted open lab sessions to facilitate hands-on learning
- Tested assignments and addressed student queries to ensure their thorough understanding of course concepts

ACADEMIC PROJECTS

SmartMedAI: Fine-tuning MedicalQA-based LLM using Reinforcement Learning from AI Feedback (RLAIF)

- Fine-tuned a MedicalQA-based LLM using LoRA, which optimized training efficiency and enabled domain-specific adaptation
- Built a preference dataset using RLAIF with a secondary LLM evaluating open-ended medical responses for DPO
- Developed a Direct Preference Optimization (DPO) pipeline to align the model for clinically accurate and relevant responses, achieving 80.4% accuracy and 0.568 BLEURT Score

SmartPDF Chat: AI-Powered Document Analysis

- Created a Python web application for natural language querying of multiple PDFs leveraging LangChain, PyPDF2, and Streamlit
- Implemented an end-to-end pipeline for text extraction, chunking, and vector representation with sentence-transformers
- Achieved precise response generation through similarity matching with faiss-cpu, flan-t5-xxl model, and InstructorEmbedding

Transfer Learning for Image Classification

- Developed a multi-class image classifier using transfer learning with ResNet50, ResNet101, EfficientNetB0, and VGG16 models
- Applied frozen feature extraction and advanced data augmentation (cropping, rotation, flipping) to address limited data
- Achieved 91.8% accuracy with EfficientNetB0, outperforming deeper models using TensorFlow and Keras frameworks

Performance Analysis of Diffusion Model for Cloud Removal from Satellite Images

- Built a model to remove clouds from satellite images using Diffusion models and U-Net Architecture and achieved 91% accuracy
- Researched 10 papers related to cloud removal techniques for literature survey and result comparison with other techniques
- Utilized Google Colab, Python, PyTorch, NumPy, and Matplotlib for implementation

Stock Analytics Web Application

- Created a responsive React UI, integrating AJAX calls for real-time stock data from Finnhub deploying Node.js and Express.js
- Implemented features dynamic Highcharts charts and social media sharing, enhancing user interaction and data visualization
- Implemented MongoDB for data management and deployed the application on AWS, specifically using Amazon EC2 instances